

IIIT, Bhubaneswar

ET 21114: Image Processing and Computer Vision

(7th Sem. ETC)

Full Marks: 30

Mid Semester Examination

Duration: 90 mins.

(Answer any three questions including Q1. All questions carry equal marks)

1. (a) How an image is represented using illumination and reflectance component? [2×5=10]
(b) What are the categories of images generated due to energy sources from EM spectrum?
(c) Generate a 5×5 Gaussian kernel using Pascal's triangle method.
(d) Explain why the discrete histogram equalization technique does not yield a flat histogram in general as the continuous histogram technique.
(e) What is chessboard distance?

2. (a) Two images of same scene is given. One of them is a reference image. What transformation technique can be used to align both the images? Explain the operations in detail. [5]

(b) What are the various adjacencies of a pixel? Let $V = \{0,1\}$ be the set of intensity values used to define adjacency. Compute the lengths of the shortest 4-, 8-, and m-path between p and q in the following image. If a particular path does not exist between these two points, explain why. [5]

3	1	2	1 (q)
2	2	0	2
1	2	1	1
(p) 1	0	1	2

3. (a) What are various order statistics filters. Explain with an example. [5]

(b) An image has gray levels ranging from 0 to 7. Plot both the histogram and normalized histogram of the image. Using the normalized value, compute the mean and standard deviation of this image. The histogram of this image is given as: [5]

Gray level	0	1	2	3	4	5	6	7
Frequency	40	45	55	50	40	35	65	70

4. (a) Explain the Laplacian operator with its application. How the background features of an image can be restored after performing Laplacian operator. [5]

(b) Explain homomorphic filtering. Also, explain which filter can be used to gain control over both the illumination and reflectance component. [5]

Ranjan